

Machine Learning

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Chapter 10: Linear latent variable models

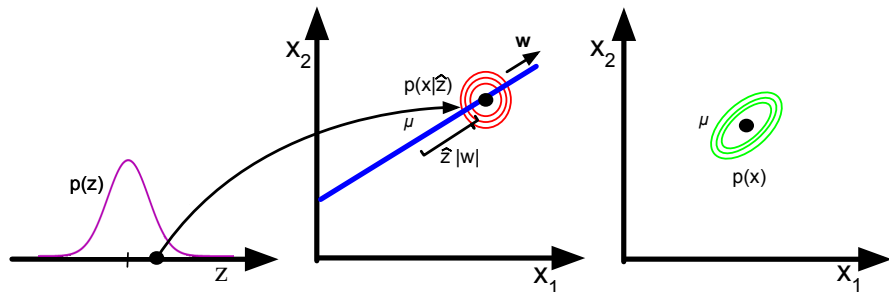
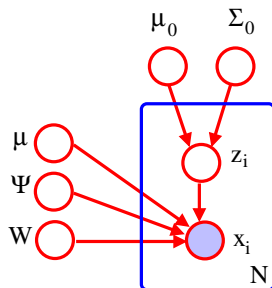


Figure 12.1 in K. Murphy: Machine Learning. MIT Press 2012.

Factor analysis

- One problem with mixture models: **only a single latent variable**. Each observation can only come from one of K prototypes.
- Alternative: $\mathbf{z}_i \in \mathbb{R}^k$. Gaussian prior:
$$p(\mathbf{z}_i) = \mathcal{N}(\mathbf{z}_i | \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$$



- For observations $\mathbf{x}_i \in \mathbb{R}^p$, we may use a **Gaussian likelihood**.
- As in linear regression, we assume the mean is a **linear** function:

$$p(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta}) = \mathcal{N}(W\mathbf{z}_i + \boldsymbol{\mu}, \boldsymbol{\Psi}),$$

W : **factor loading matrix**, and $\boldsymbol{\Psi}$: **covariance matrix**.

- We take $\boldsymbol{\Psi}$ to be **diagonal**, since the whole point of the model is to “force” $\mathbf{z}_i$ to **explain the correlation**.

Factor analysis: generative process

Generative process ($k = 1$, $p = 2$, diagonal Ψ):

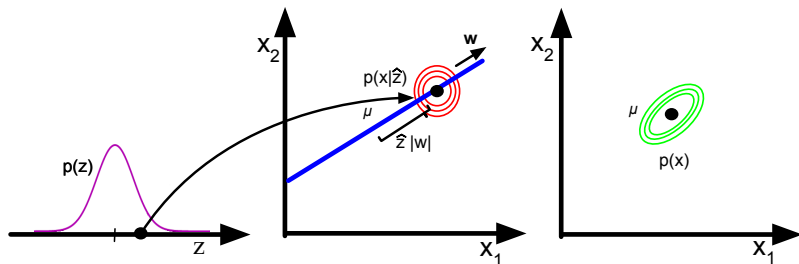


Figure 12.1 in K. Murphy: Machine Learning. MIT Press 2012.

We take an isotropic Gaussian “spray can” and slide it along the 1d line defined by $wz_i + \mu$. This induces a correlated Gaussian in 2d.

FA is a low rank parameterization of an MVN

- The induced marginal distribution $p(\mathbf{x}_i|\boldsymbol{\theta})$ is Gaussian:

$$\begin{aligned} p(\mathbf{x}_i|\boldsymbol{\theta}) &= \int \mathcal{N}(\mathbf{x}_i, W\mathbf{z}_i + \boldsymbol{\mu}, \Psi) \mathcal{N}(\mathbf{z}_i|\boldsymbol{\mu}_0, \Sigma_0) d\mathbf{z}_i, \\ &= \mathcal{N}(\mathbf{x}_i|W\boldsymbol{\mu}_0 + \boldsymbol{\mu}, \Psi + W\Sigma_0W^t) \end{aligned}$$

- We can set $\boldsymbol{\mu}_0 = \mathbf{0}$ without loss of generality (absorb $W\boldsymbol{\mu}_0$ into $\boldsymbol{\mu}$). Similarly, we can set $\Sigma_0 = I$ using $\tilde{W} = W\Sigma_0^{-1/2}$.

Covariance structure:

$$\text{cov}[\mathbf{x}|\boldsymbol{\theta}] = (W\Sigma_0^{-1/2})\Sigma_0(W\Sigma_0^{-1/2})^t + \Psi = WW^t + \Psi.$$

- We thus see that FA approximates the covariance matrix of $\mathbf{x}$ using a low-rank decomposition:

$$C := \text{cov}[\mathbf{x}] = WW^t + \Psi.$$

Only $O(p \cdot k)$ parameters $\rightsquigarrow$ compromise between full covariance with $O(p^2)$ params, and a diagonal covariance $O(p)$ params.

Inference of the latent factors

- We hope that the latent factors $\mathbf{z}$ will reveal something interesting about the data $\rightsquigarrow$ compute posterior over the latent variables:

$$\begin{aligned}p(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta}) &= \mathcal{N}(\mathbf{z}_i | \mathbf{m}_i, \Sigma) \\ \Sigma &= (I + W^t \Psi^{-1} W)^{-1} \\ \mathbf{m}_i &= \Sigma W^t \Psi^{-1} \mathbf{x}_i\end{aligned}$$

- The posterior means $\mathbf{m}_i$ are called the **latent scores, or latent factors.**

Example

- Example from (Shalizi 2009). $p = 11$ variables and $n = 387$ cases describing aspects of cars: engine size, $\#(\text{cylinders})$, miles per gallon (MPG), price, etc.
- Fit a $p = 2$ dim model. Plot m_i scores as points in $\mathbb{R}^2$.
- To get a better understanding of the “meaning” of the latent factors, project unit vectors $e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, etc. into the low dimensional space (blue lines)
- Horizontal axis represents price, corresponding to the features labeled “dealer” and “retail”, with expensive cars on the right. Vertical axis represents fuel efficiency (measured in terms of MPG) versus size: heavy vehicles are less efficient and are higher up, whereas light vehicles are more efficient and are lower down.
- Verify by finding the closest exemplars in the training set.

Example

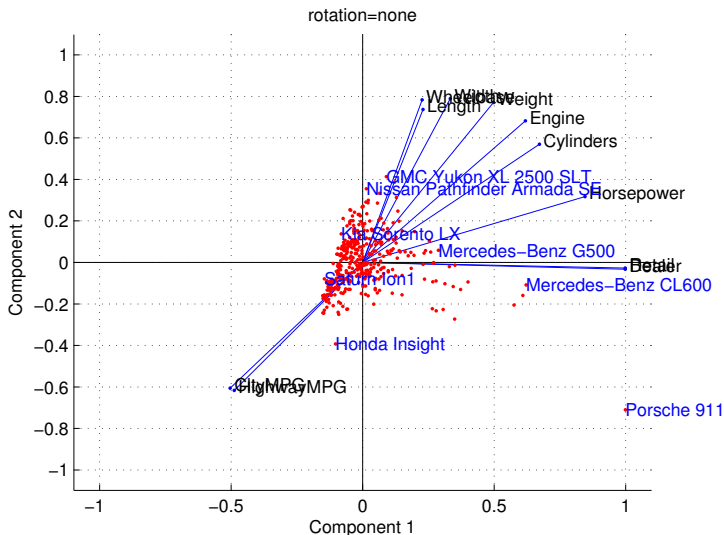
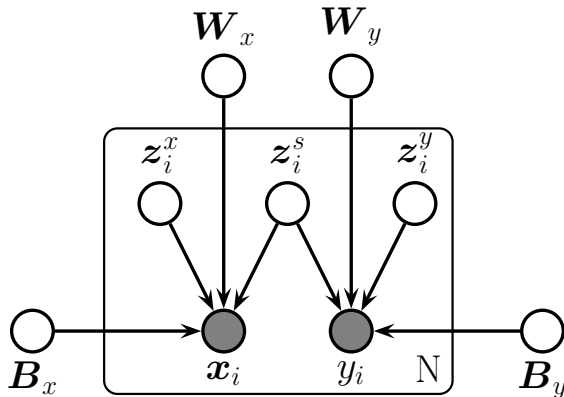


Figure 12.2 in K. Murphy: Machine Learning. MIT Press 2012.

Special Cases: PCA and CCA

- Covariance matrix $\Psi = \sigma^2 I \rightsquigarrow$ (probabilistic) **PCA**.
- Two-view version involving $\mathbf{x}$ and $\mathbf{y} \rightsquigarrow$ **CCA**.



From figure 12.19 in K. Murphy: Machine Learning. MIT Press 2012.

PCA and dimensionality reduction

Given n data points in p dimensions:

$$X = \begin{bmatrix} - & \mathbf{x}_1 & - \\ - & \mathbf{x}_2 & - \\ - & \vdots & - \\ - & \mathbf{x}_n & - \end{bmatrix} \in \mathbb{R}^{n \times p}$$

Want to reduce dimensionality from p to k . Choose k directions $\mathbf{w}_1, \dots, \mathbf{w}_k$, arrange them as columns in matrix W :

$$W = \begin{bmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \dots & \mathbf{w}_k \end{bmatrix} \in \mathbb{R}^{p \times k}$$

For each $\mathbf{w}_j$, compute **similarity** $z_j = \mathbf{w}_j^t \mathbf{x}$, $j = 1 \dots k$.

Project $\mathbf{x}$ down to $\mathbf{z} = (z_1, \dots, z_k)^t = W^t \mathbf{x}$. How to choose W ?

Encoding-decoding model

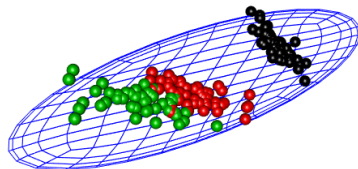
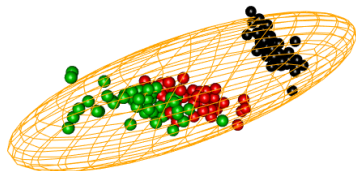
The projection matrix W serves two functions:

- **Encode:** $\mathbf{z} = W^t \mathbf{x}$, $\mathbf{z} \in \mathbb{R}^k$, $z_j = \mathbf{w}_j^t \mathbf{x}$.
 - ▶ The vectors $\mathbf{w}_j$ form a basis of the projected space.
 - ▶ We will require that this basis is orthonormal, i.e. $W^t W = I$.
- **Decode:** $\tilde{\mathbf{x}} = W \mathbf{z} = \sum_{j=1}^k z_j \mathbf{w}_j$, $\tilde{\mathbf{x}} \in \mathbb{R}^p$.
 - ▶ If $k = p$, the above orthonormality condition implies $W^t = W^{-1}$, and encoding can be undone without loss of information.
 - ▶ If $k < p$, we use the pseudo-inverse
 $\leadsto$ the reconstruction error will be nonzero.
- Above we assumed that the origin of the coordinate system is in the sample mean, i.e. $\sum_i \mathbf{x}_i = 0$.

Principal Component Analysis (PCA)

In the general case, we want the reconstruction error $\|\mathbf{x} - \tilde{\mathbf{x}}\|$ to be small.

Objective: minimize $\min_{W \in \mathbb{R}^{p \times k}: W^t W = I} \sum_{i=1}^n \|\mathbf{x}_i - WW^t \mathbf{x}_i\|^2$



Finding the principal components

Projection vectors are orthogonal $\rightsquigarrow$ can treat them separately:

$$\begin{aligned} \min_{\mathbf{w}: \|\mathbf{w}\|=1} \sum_{i=1}^n \|\mathbf{x}_i - \mathbf{w}\mathbf{w}^t \mathbf{x}_i\|^2 \\ \sum_i \|\mathbf{x}_i - \mathbf{w}\mathbf{w}^t \mathbf{x}_i\|^2 &= \sum_{i=1}^n [\mathbf{x}_i^t \mathbf{x}_i - 2\mathbf{x}_i^t \mathbf{w}\mathbf{w}^t \mathbf{x}_i + \underbrace{\mathbf{x}_i^t \mathbf{w}\mathbf{w}^t \mathbf{w}\mathbf{w}^t \mathbf{x}_i}_{=1}] \\ &= \sum_i [\mathbf{x}_i^t \mathbf{x}_i - \mathbf{x}_i^t \mathbf{w}\mathbf{w}^t \mathbf{x}_i] \\ &= \sum_i \mathbf{x}_i^t \mathbf{x}_i - \mathbf{w}^t \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^t \mathbf{w} \\ &= \underbrace{\sum_i \mathbf{x}_i^t \mathbf{x}_i}_{\text{const.}} - \mathbf{w}^t \mathbf{X}^t \mathbf{X} \mathbf{w}. \end{aligned}$$

Finding the principal components

- Want to maximize $\mathbf{w}^t X^t X \mathbf{w}$ under the constraint $\|\mathbf{w}\| = 1$
- Can also maximize the ratio $J(\mathbf{w}) = \frac{\mathbf{w}^t X^t X \mathbf{w}}{\mathbf{w}^t \mathbf{w}}$ and rescale $\hat{\mathbf{w}}$.
- Optimal projection u is the eigenvector of $X^t X$ with largest eigenvalue (J is a Rayleigh quotient).
- Note that we assumed that $\sum_i \mathbf{x}_i = 0$. Thus, the columns of X are assumed to sum to zero.
 - ↪ compute SVD of “centered” matrix X
 - ↪ column vectors in W are eigenvectors of $X^t X$
 - ↪ they are the principal components.

Eigen-faces [Turk and Pentland, 1991]

- p = number of pixels
- Each $\mathbf{x}_i \in \mathbb{R}^p$ is a face image
- x_{ji} = intensity of the j -th pixel in image i

$$(X^t)_{p \times n} \approx W_{p \times k} (Z^t)_{k \times n}$$

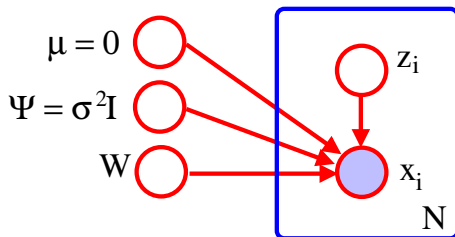
 $\approx$  $\begin{bmatrix} | & & | \\ \mathbf{z}_1 & \dots & \mathbf{z}_n \\ | & & | \end{bmatrix}$

Idea: $\mathbf{z}_i$ more 'meaningful' representation of i -th face than $\mathbf{x}_i$

Can use $\mathbf{z}_i$ for nearest-neighbor classification

Much faster when $p \gg k$.

Probabilistic PCA



- Assuming $\Psi = \sigma^2 I$ and centered data in the FA model
 $\rightsquigarrow$ likelihood and marginal likelihood

$$p(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta}) = \mathcal{N}(\mathbf{W} \mathbf{z}_i, \sigma^2 I),$$

$$\begin{aligned} p(\mathbf{x}_i | \boldsymbol{\theta}) &= \int \mathcal{N}(\mathbf{x}_i, \mathbf{W} \mathbf{z}_i, \sigma^2 I) \mathcal{N}(\mathbf{z}_i | \mathbf{0}, I) d\mathbf{z}_i, \\ &= \mathcal{N}(\mathbf{x}_i | \mathbf{0}, \sigma^2 I + \mathbf{W} \mathbf{W}^t) \end{aligned}$$

Probabilistic PCA

- (Tipping & Bishop 1999): Maxima of the marg. likelihood given by

$$\hat{W} = V(\Lambda - \sigma^2 I)^{\frac{1}{2}} R,$$

where R is an arbitrary orthogonal matrix,
columns of V : first k eigenvectors of $S = \frac{1}{n} X^t X$,
 Λ : diagonal matrix of eigenvalues.

- As $\sigma^2 \rightarrow 0$, we have $\hat{W} \rightarrow V$, **as in classical PCA** (for $R = \Lambda^{-\frac{1}{2}}$).
- Projections $\mathbf{z}_i$: Posterior over the latent factors:

$$p(\mathbf{z}_i | \mathbf{x}_i, \hat{\boldsymbol{\theta}}) = \mathcal{N}(\mathbf{z}_i | \hat{\mathbf{m}}_i, \sigma^2 \hat{\mathbf{F}}^{-1})$$

$$\hat{\mathbf{F}} = \sigma^2 I + \hat{W}^t \hat{W}$$

$$\mathbf{m}_i = \hat{\mathbf{F}}^{-1} \hat{W}^t \mathbf{x}_i$$

For $\sigma^2 \rightarrow 0$, $\mathbf{z}_i \rightarrow \mathbf{m}_i$ and $\mathbf{m}_i \rightarrow V^t \mathbf{x}_i \rightsquigarrow$ orthogonal projection of the data onto the column space of V , **as in classical PCA**.